Interacting Multipoles of Rank 1 and 1:

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \delta\left(\alpha,\beta\right) \cdot \mu_\alpha \cdot \mu_\beta \end{bmatrix} \tag{1}$$

Simplify deltas

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \delta\left(\alpha,\alpha\right) \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \tag{2a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \mu_\beta \\ -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \tag{2b}$$

Simplify R

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_z \cdot R_z \cdot \mu_z \cdot \mu_z \\ -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_\alpha \cdot \mu_\alpha \end{bmatrix} \tag{3b}$$

Simplify Multipoles

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{4b}$$

Exploit Traceless Conditions

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{5a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{5b}$$

Reduce Indices

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{6b}$$

Add up

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y \cdot \mu_y \end{bmatrix} \tag{7}$$

Combining everything

$$T_{\alpha\beta}\mu_\alpha\mu_\beta = \left[\ +1 \cdot R_z^{-3} \cdot \mu_y \cdot \mu_y\ \right] \tag{8}$$